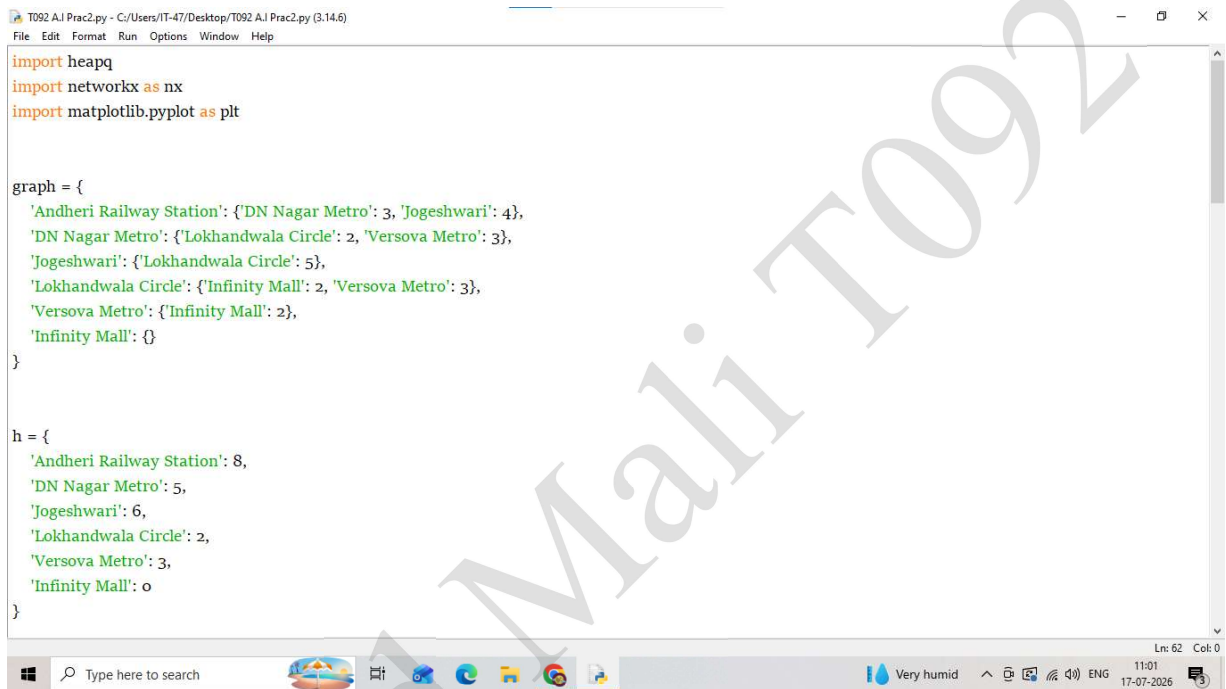


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Practical No 2

Aim :-

- Implement the A* Search algorithm for solving a pathfinding problem.
- Implement the Recursive Best-First Search algorithm for the same problem.
- Compare the performance and effectiveness of both algorithms.

Code :- A* Algorithm



```
T092 A.I Prac2.py - C:/Users/IT-47/Desktop/T092 A.I Prac2.py (3.14.6)
File Edit Format Run Options Window Help

import heapq
import networkx as nx
import matplotlib.pyplot as plt

graph = {
    'Andheri Railway Station': {'DN Nagar Metro': 3, 'Jogeshwari': 4},
    'DN Nagar Metro': {'Lokhandwala Circle': 2, 'Versova Metro': 3},
    'Jogeshwari': {'Lokhandwala Circle': 5},
    'Lokhandwala Circle': {'Infinity Mall': 2, 'Versova Metro': 3},
    'Versova Metro': {'Infinity Mall': 2},
    'Infinity Mall': {}
}

h = {
    'Andheri Railway Station': 8,
    'DN Nagar Metro': 5,
    'Jogeshwari': 6,
    'Lokhandwala Circle': 2,
    'Versova Metro': 3,
    'Infinity Mall': 0
}
```

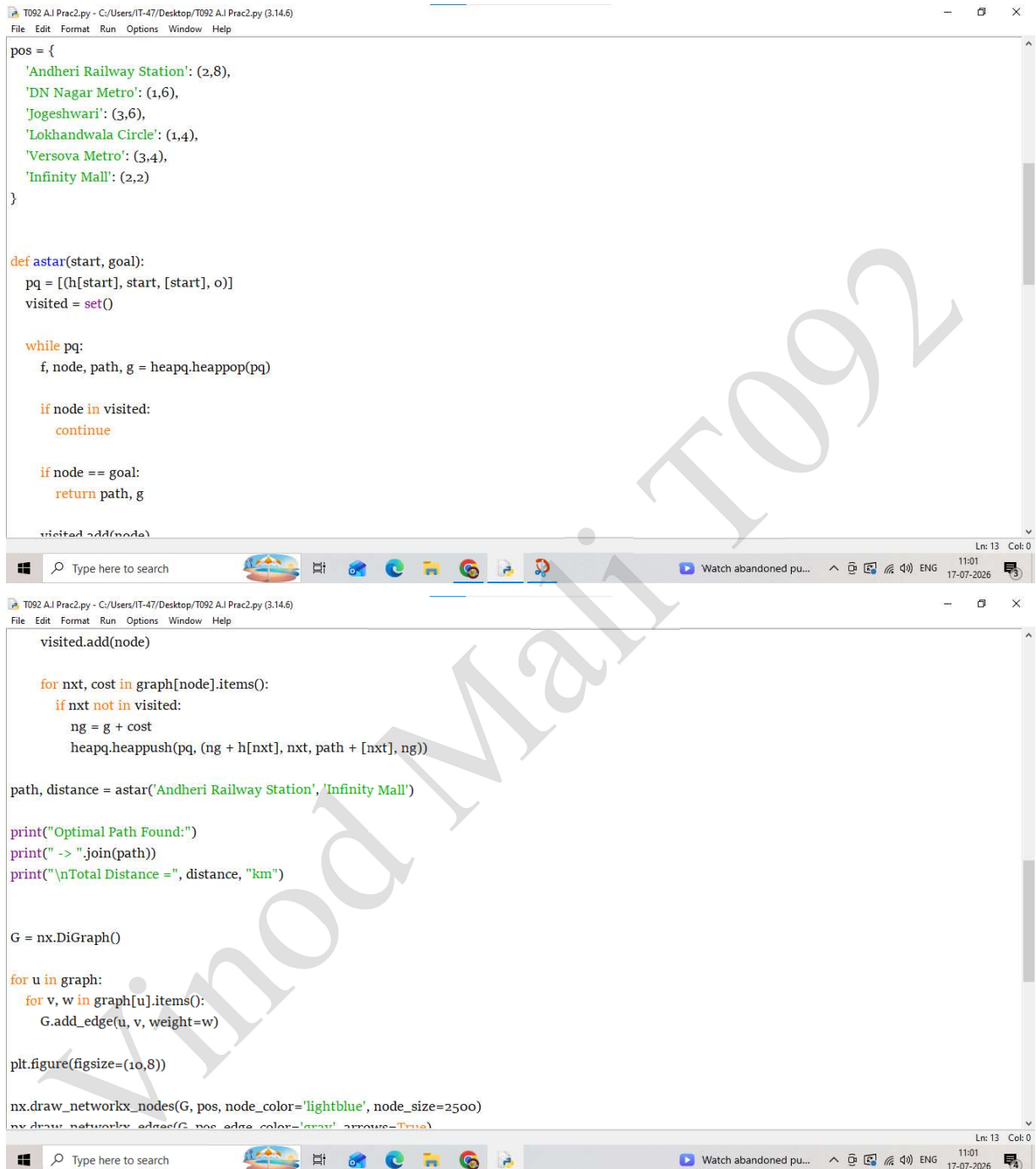
The screenshot shows a Python script in a code editor. The script defines a graph with six nodes and their weighted edges, and a heuristic function 'h' for each node. The graph is an undirected graph with the following edges and weights: (Andheri Railway Station, DN Nagar Metro) with weight 3, (Andheri Railway Station, Jogeshwari) with weight 4, (DN Nagar Metro, Lokhandwala Circle) with weight 2, (DN Nagar Metro, Versova Metro) with weight 3, (Jogeshwari, Lokhandwala Circle) with weight 5, (Lokhandwala Circle, Infinity Mall) with weight 2, (Versova Metro, Infinity Mall) with weight 2. The heuristic function 'h' assigns values: Andheri Railway Station: 8, DN Nagar Metro: 5, Jogeshwari: 6, Lokhandwala Circle: 2, Versova Metro: 3, Infinity Mall: 0. The code imports heapq, networkx as nx, and matplotlib.pyplot as plt.

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```

T092 A.I Prac2.py - C:/Users/IT-47/Desktop/T092 A.I Prac2.py (3.14.6)
File Edit Format Run Options Window Help

pos = {
    'Andheri Railway Station': (2,8),
    'DN Nagar Metro': (1,6),
    'Jogeshwari': (3,6),
    'Lokhandwala Circle': (1,4),
    'Versova Metro': (3,4),
    'Infinity Mall': (2,2)
}

def astar(start, goal):
    pq = [(h[start], start, [start], 0)]
    visited = set()

    while pq:
        f, node, path, g = heapq.heappop(pq)

        if node in visited:
            continue

        if node == goal:
            return path, g

        visited.add(node)

        for nxt, cost in graph[node].items():
            if nxt not in visited:
                ng = g + cost
                heapq.heappush(pq, (ng + h[nxt], nxt, path + [nxt], ng))

path, distance = astar('Andheri Railway Station', 'Infinity Mall')

print("Optimal Path Found:")
print(" -> ".join(path))
print("\nTotal Distance = ", distance, "km")

G = nx.DiGraph()

for u in graph:
    for v, w in graph[u].items():
        G.add_edge(u, v, weight=w)

plt.figure(figsize=(10,8))

nx.draw_networkx_nodes(G, pos, node_color='lightblue', node_size=2500)
nx.draw_networkx_edges(G, pos, edge_color='gray', arrows=True)

```

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```
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File Edit Format Run Options Window Help

nx.draw_networkx_nodes(G, pos, node_color='lightblue', node_size=2500)
nx.draw_networkx_edges(G, pos, edge_color='gray', arrows=True)

nx.draw_networkx_edges(
    G,
    pos,
    edgelist=list(zip(path, path[1:])),
    edge_color='orange',
    width=4,
    arrows=True
)

nx.draw_networkx_labels(
    G,
    pos,
    {n: f'{n}\nh(n)={h[n]}' for n in graph},
    font_size=9,
    font_weight='bold'
)

nx.draw_networkx_edge_labels(
    G,
    pos,
    {(u, v): f'{w} km' for u, v, w in G.edges.data('weight')},
    font_color='red'
)

plt.title("A* Search Route\nAndheri Railway Station to Infinity Mall")
plt.axis("off")
plt.show()
```

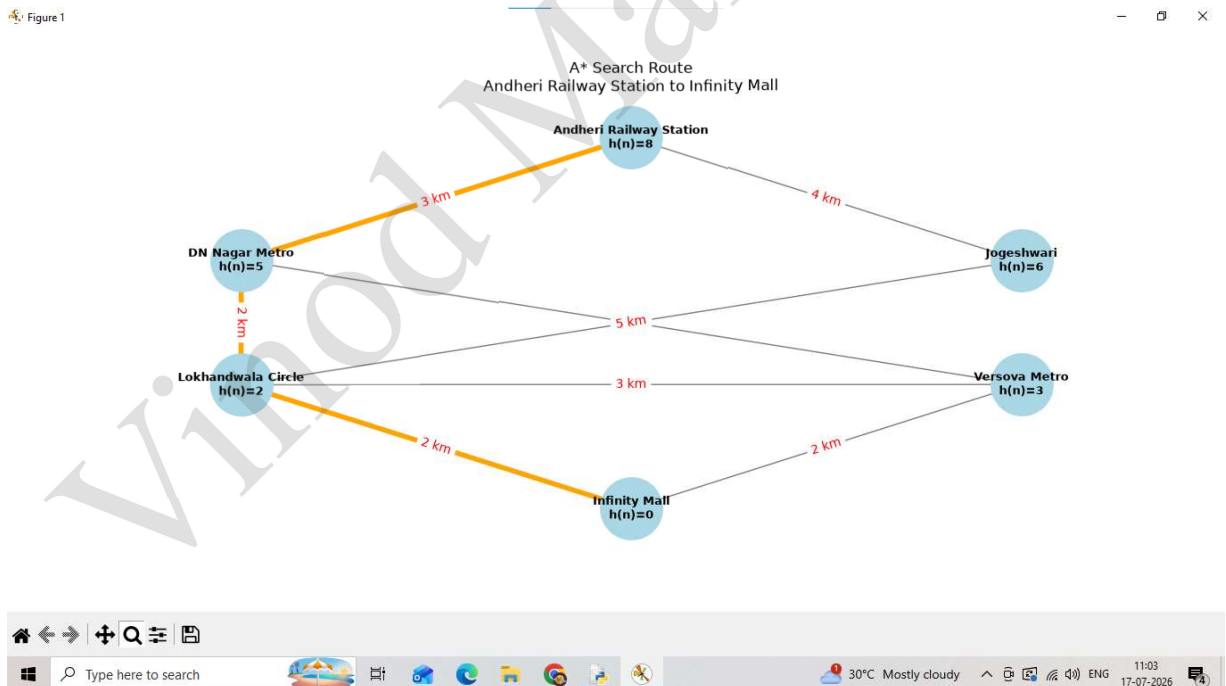
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OutPut :-

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:/Users/IT-47/Desktop/To92 A.I Prac2.py =====
Optimal Path Found:
Andheri Railway Station -> DN Nagar Metro -> Lokhandwala Circle -> Infinity Mall

Total Distance = 7 km
```



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Code :- Recursive Best-First Search algorithm

```
T092 A.I Prac2.py - C:/Users/IT-47/Desktop/T092 A.I Prac2.py (3.14.6)
File Edit Format Run Options Window Help

import networkx as nx
import matplotlib.pyplot as plt

|
graph = {
    'Andheri Railway Station': {'Bandra': 10, 'Jogeshwari': 4},
    'Jogeshwari': {'Bandra': 8},
    'Bandra': {'Dadar': 9},
    'Dadar': {'Byculla': 8},
    'Byculla': {'CST': 5},
    'CST': {}
}

h = {
    'Andheri Railway Station': 24,
    'Jogeshwari': 22,
    'Bandra': 14,
    'Dadar': 8,
    'Byculla': 4,
    'CST': 0
}

Ln: 107 Col: 0

T092 A.I Prac2.py - C:/Users/IT-47/Desktop/T092 A.I Prac2.py (3.14.6)
File Edit Format Run Options Window Help

pos = {
    'Andheri Railway Station': (1,8),
    'Jogeshwari': (3,7),
    'Bandra': (2,5),
    'Dadar': (3,3),
    'Byculla': (4,2),
    'CST': (5,1)
}

def rbfs(node, goal, g, limit, path):

    if node == goal:
        return path, g, g

    s = [[n, g+c, max(g+c+h[n], g+h[node])]]
    for n, c in graph[node].items():

    if not s:
        return None, float('inf'), float('inf')

    while True:
        s.sort(key=lambda x: x[2])

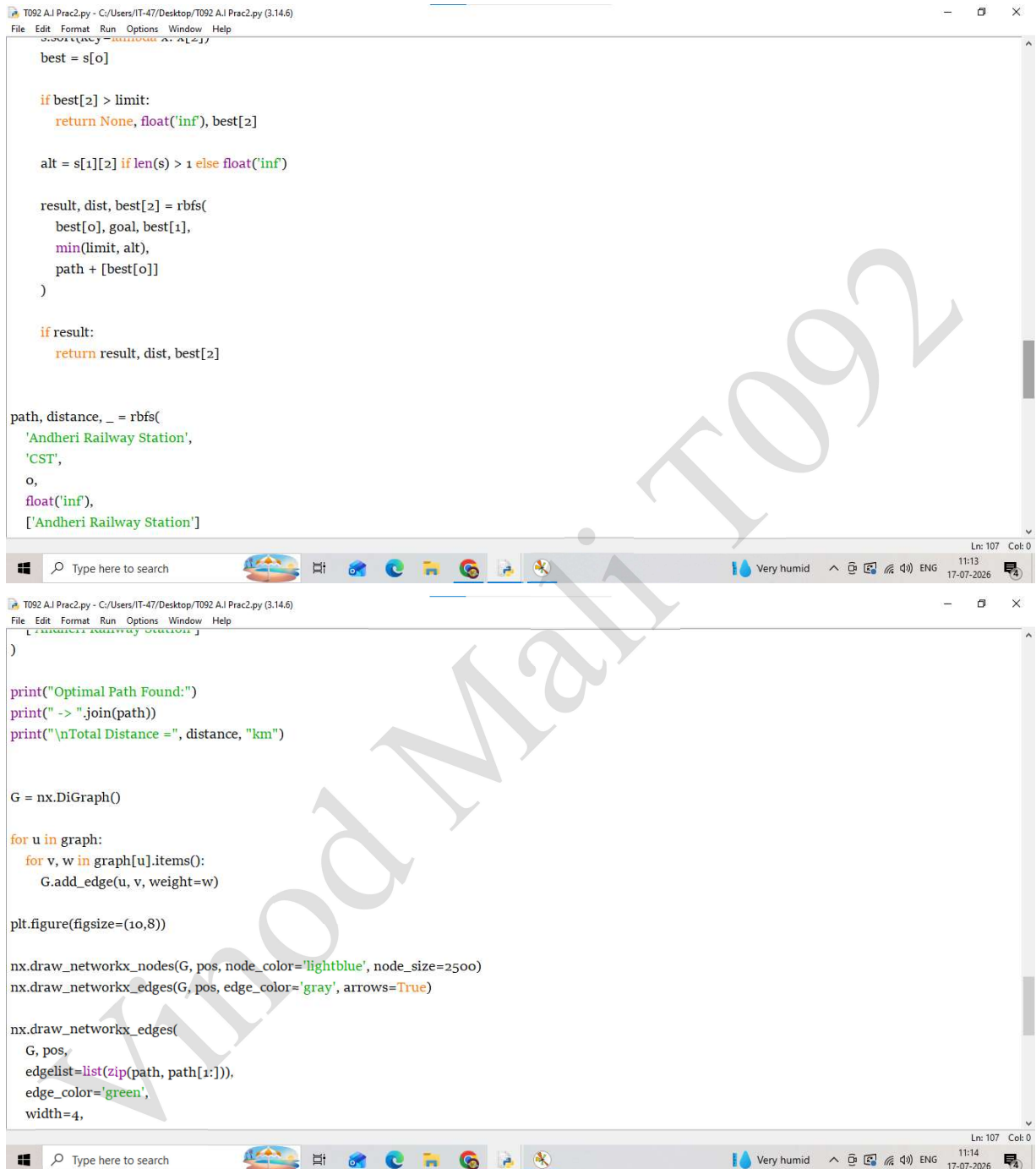
Ln: 107 Col: 0
```

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```

best = s[o]

if best[2] > limit:
    return None, float('inf'), best[2]

alt = s[1][2] if len(s) > 1 else float('inf')

result, dist, best[2] = rbfs(
    best[o], goal, best[1],
    min(limit, alt),
    path + [best[o]]
)

if result:
    return result, dist, best[2]

path, distance, _ = rbfs(
    'Andheri Railway Station',
    'CST',
    o,
    float('inf'),
    ['Andheri Railway Station']
)

print("Optimal Path Found:")
print("-> ".join(path))
print("\nTotal Distance =", distance, "km")

G = nx.DiGraph()

for u in graph:
    for v, w in graph[u].items():
        G.add_edge(u, v, weight=w)

plt.figure(figsize=(10,8))

nx.draw_networkx_nodes(G, pos, node_color='lightblue', node_size=2500)
nx.draw_networkx_edges(G, pos, edge_color='gray', arrows=True)

nx.draw_networkx_edges(
    G, pos,
    edgelist=list(zip(path, path[1:])),
    edge_color='green',
    width=4,

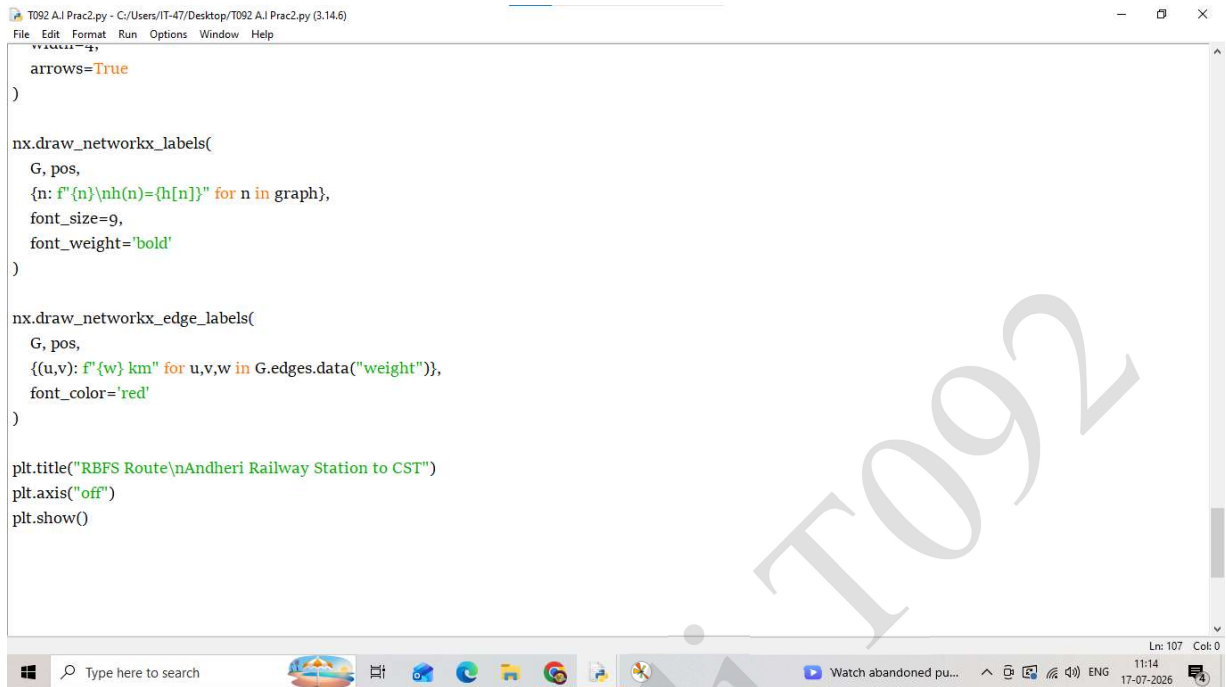
```

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```

arrows=True
)

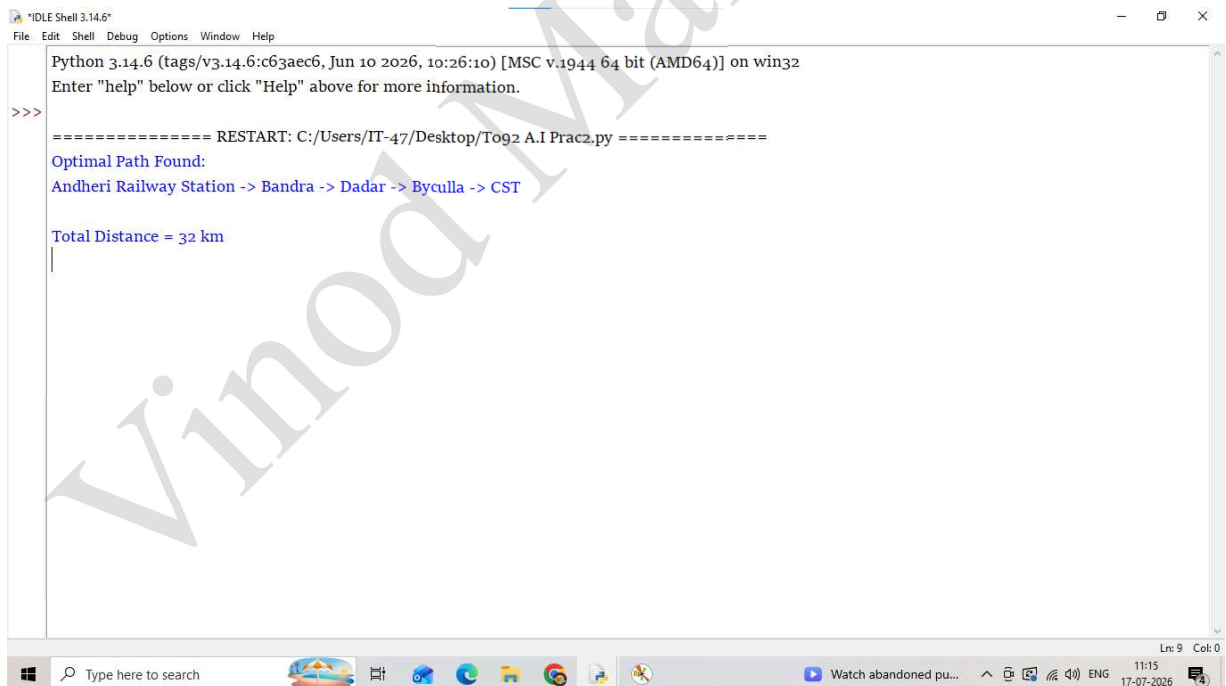
nx.draw_networkx_labels(
    G, pos,
    {n: f'{n}\nh(n)={h[n]}' for n in graph},
    font_size=9,
    font_weight='bold'
)

nx.draw_networkx_edge_labels(
    G, pos,
    {(u,v): f'{w} km' for u,v,w in G.edges.data("weight")},
    font_color='red'
)

plt.title("RBFS Route\nAndheri Railway Station to CST")
plt.axis("off")
plt.show()

```

OutPut :-



```

Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/IT-47/Desktop/To92 A.I Prac2.py =====
Optimal Path Found:
Andheri Railway Station -> Bandra -> Dadar -> Byculla -> CST

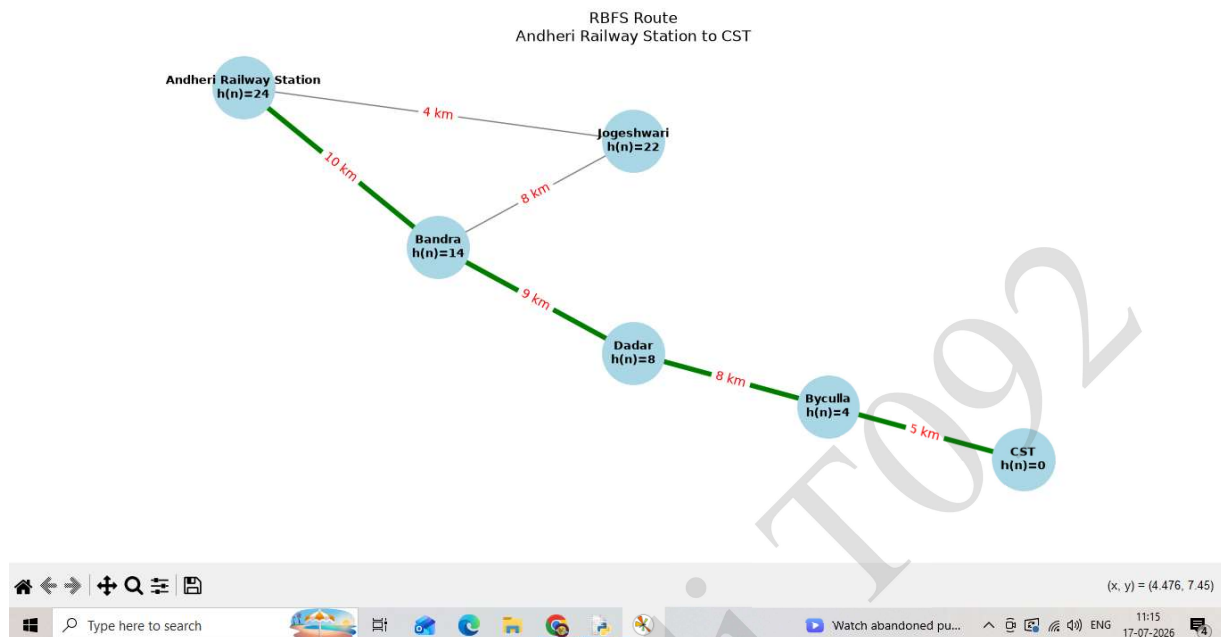
Total Distance = 32 km

```

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Figure 1



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Compare the performance and effectiveness of both algorithms.

Metric	A Search*	Recursive Best-First Search (RBFS)
Search Method	Uses a Priority Queue	Uses Recursion
Evaluation Function	$f(n) = g(n) + h(n)$	$f(n) = \max(g(n) + h(n), \text{parent's } f(n))$
Memory Usage	High	Low
Time Complexity	$O(bd)$ (Generally faster)	$O(bd)$ (May be slower due to backtracking)
Space Complexity	$O(bd)$ (Stores all generated nodes)	$O(bd)$ (Stores only the current search path)
Completeness	Yes	Yes
Optimality	Yes (with an admissible heuristic)	Yes (with an admissible heuristic)
Execution Speed	Faster	Slightly Slower
Backtracking	Not Required	Required
Best Suitable For	Fast pathfinding applications	Memory-constrained systems
Route Found	Andheri Railway Station → Bandra → Dadar → Byculla → CST	Andheri Railway Station → Bandra → Dadar → Byculla → CST
Total Distance	32 km	32 km

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